

Anuprovo Debnath

Fourth-year Undergraduate (BS-MS), Physics

✉ ad23ms110@iiserkol.ac.in ☎ 062896 30036 📍 Kolkata, India 🌐 anuprovo-debnath.github.io

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Education

BS-MS Indian Institute of Science Education and Research, Kolkata, Physics

Kalyani, India

Aug 2023 – present

- CGPA: 8.82
- Major in Physics; Minor in Mathematics and Computer Science
- Coursework: Quantum Physics, Statistical Mechanics, Computational Physics, Stochastic Processes, Mathematical Methods, Electrodynamics, Nuclear Physics Lab

St. Jude's High School, ISC (Class XII)

Kolkata, India

2021 – 2023

- Score: 95.7%

St. Jude's High School, ICSE (Class X)

Kolkata, India

2020 – 2021

- Score: 94.4%

Experience

Institute for Plasma Research (IPR), Research Student (Guide: Prof. Sudip Sengupta)

Gandhinagar, India

Numerical Modeling of Laser-Plasma Wakefield Acceleration in the Basic Theory and Simulation group.

May 2026 – present

5 months

- Developed a Python-based RK4 computational framework to model electron trapping and acceleration in laser-driven plasma waves
- Verified test electron energy gain in longitudinal sinusoidal electric fields, reproducing Tajima and Dawson (1979) scalings
- Extended model to 3D with perpendicular magnetic fields to simulate Surfatron acceleration regime
- Developing a custom 1D Electrostatic Particle-in-Cell (PIC) simulation code from scratch for kinetic instabilities

CSIR - Central Mechanical Engineering Research Institute, IASc Summer Research Fellow (Guide: Dr. T. Murugan)

Durgapur, India

Simulation and Analysis of ns-DBD Plasma Actuator Shock Waves (SRFP 2026).

June 2026 – July 2026

2 months

- Engineered OpenFOAM framework to simulate ns-DBD plasma actuator shock waves at 30, 70, and 100 kPa
- Implemented 100 ns phenomenological energy deposition model on multi-block grid, bypassing plasma chemistry overhead
- Developed Python/PyVista pipeline using Modified Friedlander Waveform fitting and Levenberg-Marquardt uncertainty propagation
- Validated numerical shock propagation against benchmark experimental data
- [View 8-Week SRFP Project Report](#)

Physical Research Laboratory (PRL), Ahmedabad , Research Intern (Guide: Dr. Prashant Kumar) Evaluating Plasma Parameters from a Typical LIBS Plasma. - Developed Python framework for extracting Te and ne from LIBS plasmas using Boltzmann plots, Saha-Boltzmann analysis, and Stark broadening with NIST data - Demonstrated reliable plasma parameter estimation using calibration-free LIBS - Implemented machine-learning framework with LTE validation, self-absorption modeling, and spectral line analysis View LIBS Summer Internship Report	Ahmedabad, India May 2025 – Oct 2025 6 months
IISER Kolkata , Student Researcher (Guide: Dr. Sudip K. Garain) Simulating accretion disk dynamics around black holes in a relativistic framework. Relativistic fluid dynamics modeling and simulations	Kolkata, India Feb 2025 – present 1 year 8 months
IISER Kolkata , Reading Project (Guide: Prof. Arindam Kundagrami) Studied thermodynamics and statistical mechanics following ‘A Treatise on Heat’ by M. N. Saha and B. N. Srivastava.	Kolkata, India May 2024 – July 2024 3 months

Leadership and Activities

Slashdot (Coding & Design Club, IISER Kolkata) , Secretary Leading technical and organizational initiatives for the coding and designing club during 2025–26 tenure. - Setup of a hobbyist compute cluster using refurbished PCs with Proxmox VE and Kubernetes - Organized IBM-sponsored Qiskit Fall Fest 2025 (quantum computing event with hands-on learning) - Conducted Blender Workshop on Scientific Visualisation using Python scripting	IISER Kolkata Nov 2025 – present
Academic Workshops & Initiatives , Participant / Contributor - IISER Mohali Astronomy & Astrophysics Summer School (2024): Hands-on astronomy research techniques - NASA SAGAN Workshop (2024): Exoplanet Science Institute-hosted workshop on exoplanet research - iGEM Team, IISER Kolkata (2024): Contributed to website development, visualisation, and animation	India 2024 – 2024

Awards and Honors

Summer Research Fellowship Programme (SRFP) Selected for research in Physics at CSIR-CMERI by IAS / INSA / NASI	2026
Smart India Hackathon (SIH) Team ‘Haklu El At’ shortlisted for problem statement SIH25173	2025
International Astronomy and Astrophysics Competition (IAAC) Received Silver Honour	2024
DST INSPIRE Scholarship Awarded by DST, Govt. of India for ranking in the top 1% in the ISC Examination	2023
National Entrance Screening Test (NEST) All India Rank 48	2023
JBNSTS Junior Scholarship Jagadis Bose National Science Talent Search Junior Scholar	2021

KVPY Aptitude Test (Stream - SA)

2021

- All India Rank 2104

Swarnakana Interschool Merit Test

2016 - 2019

- Ranked in every test consecutively from 2016 to 2019

Skills

Programming & Simulation: Python, C, C++, Java, FORTRAN, OpenFOAM, PIC, LaTeX, Typst

Data Visualisation & Diagnostics: ParaView, PyVista, Matplotlib, ORIGIN, gnuplot

Systems & Infrastructure: Linux, Proxmox VE, Kubernetes, Git, Blender

Languages: English (Fluent), Bengali (Native), Hindi (Fluent)

Extracurriculars

- Secretary of Slashdot, the coding and designing club of IISER-K (2025–26 tenure).
- Active member of Singularity (astronomy club) and video editor for the podcast 'Talking to the Stars'.
- Gold medal-winning drama team (light & sound production) at the Inter IISER Cultural Meet (IICM 2024 & 2025).
- Assisted Prof. Rajesh Kumble Nayak (DPS, IISER-K) in setting up a hobbyist electronics lab.
- Passionate about playing guitar, reading, and listening to music.